

WORKING PAPER

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COMPUTABILITY • FORMAL EPISTEMOLOGY • ASTROBIOLOGY

Rungs, Voids and Effective Closure:

Synopsis of Programme 1 of *Effective Horizons and Cognitive Spaces*

what a mind can reach, in what time, and who can tell

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Formalisation developed with Claude, an AI system by Anthropic (see Author contributions)

Abstract. *Effective Horizons and Cognitive Spaces* is a study of what a mind can reach, in what time, and who can tell. Its first programme, *Rungs, Voids and Effective Closure*, is four working papers [1, 2, 3, 4] that develop one idea: that the reach of a cognitive agent can be modelled as a Turing ideal, that the problems an agent can approach but not settle are the limits it can effectively generate, and that iterating this *effective closure* produces a ladder of rungs with voids between them. The papers are written to stand alone and will be found separately; this note says what the idea is, where it came from, what each paper contributes, how the four fit together, and in what order to read them. It is informal by design.

WHERE IT CAME FROM

The programme began with Fermi's question and Liu's answer to it. The dark forest says the sky is silent because everyone is hiding. That is a claim about strategy, and it assumes something stronger that nobody states: that if a civilisation were not hiding we could understand what it sent. The question that started this work was whether that assumption is true, and if not, whether the failure could be made precise. What would it mean for a mind to be unable, not merely slow or under-resourced but *unable*, to reach a piece of content? Is there a difference in kind between a problem that is hard and a problem that lies in a void?

Computability is the right level for that question, because voids are about limits rather than costs. A cognitive agent settles some class of problems; whatever it can compute from what it settles, it also settles, so its reach is a Turing ideal. The one further thing an agent can do is approximate: generate a sequence of guesses it cannot certify and commit to the limit. The limits an agent can effectively generate from its own ideal form its *effective closure*, and the closure of the computable sets is exactly one jump up. Iterating gives the finite levels of the arithmetical hierarchy, and nothing else: the rungs. The ladder exists only because convergence is uncertified; if a modulus of convergence were required, the closure would be idempotent and there would be one rung. That observation, that the ladder is a consequence of the absence of certificates, is the hinge of the whole programme.

THE FOUR PAPERS

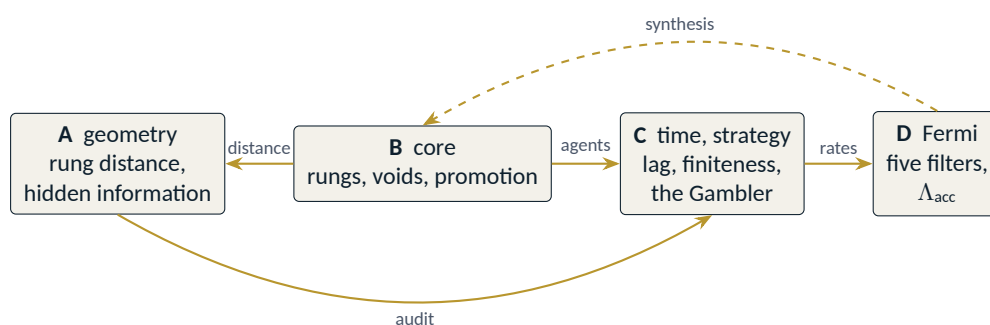
Paper B, the core [1]. Read this first. It sets out the postulates (reachability, targets as decision problems, effective approach), proves the Rung Theorem and the Stall Theorem (the union of the rungs is a fixed point and true arithmetic lies outside it), shows that voids are generic (the closure of any countable space is meagre and null), and shows that a leap across a void is unrecognisable from below: no observer of any computational power can recognise, even in the limit, that a process has left the closure of a countable space. It then says who turns a limit into content. An agent carries a certified space and a committed one, and climbs only by *promotion*, an uncertified commitment to its own limits. Content crosses between agents as method, as testimony, or as certification from strictly above; subsumption of a peer is verifiable in the limit, strict superiority never is. Its last section is the synthesis of the series: three orderings on the same objects, by reachability, by geometry and by certification, and one invariant, *limits determine what may become true; moduli and channels determine what can become usable*. How to read it: the postulates and the Rung and Stall Theorems are the content; the agent model is where the later papers attach; the Rook's Tree postscript is an image, not an argument.

Paper A, the geometry [2]. The rung distance $d(x, y) = \max(\delta(x, y), \delta(y, x))$, with $\delta(x, y)$ the number of jumps of x needed to compute y , makes the arithmetical degrees a metric space, and the paper asks what it looks like. Every canonical ascent is a geodesic ray and all rays point at one ideal point; a shadow ultrametric turns the space into a tree up to an excess that vanishes exactly on the generalised-low degrees; on a sheet spanned by two mutually random degrees the distance has a closed form; the space is not hyperbolic and not a graph metric; and the central theorem is that *hidden information is unbounded*: there are points whose first jump conceals arbitrarily much while their horofunction level is zero, built by coding a generic set with delays drawn from independent generic columns. Tree-likeness is reduced to Manning’s bottleneck property for the unit-step graph and refuted conditionally, under an open embedding hypothesis for jump partial orders. How to read it: a logician can read it on its own; everyone else needs only the first two sections and the statement of the hidden-information theorem, which is what Paper C uses.

Paper C, time and strategy [3]. The population is run forward. Under canonical ascent there is no lateral drift, only vertical recession at the rate difference between agents, and it is one-sided: the faster sees the slower. A cosmological constant makes total computation finite, so every physical agent is at rung zero for ever, and the paper defines the *settled profile* as the resource-bounded rung. The strategic part gives the information structure of the game: no universal detector of superiority, bluff-proof detectors that convince only a meagre class, randomness that demonstrates superiority in the limit while genericity cannot, subsumption demonstrable with a witness or from one rung up but by no protocol at the prover’s own rung, and, under two postulates, the sterility of conquest: the only certified growth is by consent. The *Gambler*, an agent playing from a source it does not hold, is unpredictable even when cognitively subsumed; using Paper A, its random source can be audited in hindsight three rungs up but never predicted, and a delay-coded source can be arranged to be unreadable at any prescribed rung. How to read it: theorems and postulate-consequences are marked; the physical readings (spectral gaps as certificates, horizons as purchased certificates) are suggestive, not results.

Paper D, Fermi [4]. The dynamics of C is given numbers. A rate postulate yields a contact cone, a closed-form contact probability and a *Fermi function*, the probability that an observer parses no peer within a budget; for short budgets it obeys $-\ln \mathcal{F}_N \simeq N(B/W)C$, so abundance and head start enter together as an exposure. A marked birth-and-contact process generated from cosmic history produces a directed contact graph, and sensitivity analysis and a phase diagram show that persistence of information, not bandwidth, governs connectivity, with a crossover near $L^{\text{info}}/\bar{a} \approx 0.4$ that sharpens with population. Separating an active channel (testimony) from a passive one (observability of a record) shows that only closing both conceals. The paper ends on a structure: between a physical population and an observed one lie five filters, existence, persistence, causal reach, channel access and cognitive access, which compose into an accessible intensity Λ_{acc} with $\Pr[\text{silence}] = e^{-\Lambda_{\text{acc}}}$; each darkness is the failure of one filter, and Liu’s is one of four. How to read it: as a model paper whose inputs are scenarios and whose outputs are sensitivities, with its two self-corrections (a spurious silence floor; height conflated with access) kept in the text as findings.

HOW THEY FIT



B is the hub and the other three attach at branching points it lists. A measures the structure B defines; C sets it in time and asks who can tell what about whom; D counts. One distinction runs through all four, between information that *exists* and information that is *effectively available* to a particular observer, and each paper names a different obstruction between the two: limits without certificates (B), distance and hidden depth (A), lag, finiteness and concealment (C), persistence and channels (D). Read in order they say: there is a ladder; it has a shape; climbing it takes time and can be hidden; and that is why the sky can be silent without being empty.

HOW THE IDEAS EVOLVED

The papers were iterated against external critical readings, and three retractions are kept in the text because they are informative. A lemma in Paper A asserting that excess is decided at the first jump was false, and its replacement, the sandwich $e_\infty \leq c_\infty \leq \varepsilon \leq h(x') - 1$, led directly to the hidden-information theorem. Paper D's first phase diagram reported a "silence floor of one half" that was the fraction of lineages not yet born; conditioning on the existing population removed it and sharpened the persistence crossover. And Paper D's first statement about concealment, that switching testimony off changes nothing, came from a model in which being above a peer was treated as having access to it; separating height from access produced the active/passive channel result and the fifth filter. The method, a formal core with its load-bearing assumptions flagged, consequences marked as theorem or postulate, and every correction left visible, is as much the programme's contribution as any single result.

READING ORDER, AND WHAT COMES NEXT

B, then whichever of A, C and D answers the reader's question: A for the mathematics of the rung distance, C for populations and strategy, D for Fermi. Non-logicians can skip A on first reading. Each paper's open problems are listed at its end and are meant to be taken up; the models of C and D are frozen at this version so that others can criticise them as stated. The study's title names its three questions, what a mind can reach (B, A), in what time (C) and who can tell (C, D); this programme answers the first in full and the other two in a first form, and the open problems of the four papers are the agenda of the next.

AUTHOR CONTRIBUTIONS

The programme, its questions and its central ideas (cognitive spaces as Turing ideals, effective closure, rungs and voids, promotion, the Gambler, cosmic birth times, the Fermi reading) are the author's. The formalisation, proofs, simulations and drafting were developed with Claude, an AI system by Anthropic, under the author's direction, with each paper's own contributions statement giving the detail. External critical readings of successive drafts are acknowledged in the papers.

REFERENCES

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- [3] N. Goodman. Populations and strategy in cognitive spaces: lag, horizons, finite computation and the Gambler. Working paper (Paper C), 2026.
- [4] N. Goodman. Cognitive accessibility and the Fermi problem: a rate model, a Fermi function and a marked birth-and-contact process. Working paper (Paper D), 2026.